

Instructions

- All questions carry equal weightage.
- Copying will result in a loss of marks/penalty.
- All assumptions must be clearly stated along with appropriate figures (if any); otherwise, grading will not be done.

1. The space between the plates of a parallel-plate capacitor is filled with dielectric material whose dielectric constant varies linearly from 1 at the bottom plate ($x = 0$) to 2 at the top plate ($x = d$). The capacitor is connected to a battery of voltage V . Find all the bound charge, and check that the total is zero.
2. An electric dipole p , pointing in the y direction, is placed midway between two large conducting plates, as shown in Fig. 1. Each plate makes a small angle θ with respect to the x axis, and they are maintained at potentials $\pm V$. What is the direction of the net force on p ? Draw the diagram to justify your answer.

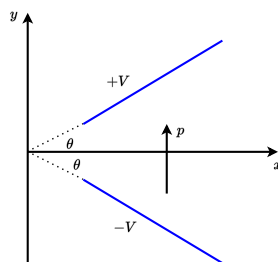


Figure 1: Q2

3. The space between the plates of a parallel-plate capacitor is filled with two slabs of linear dielectric material. Each slab has thickness a , so the total distance between the plates is $2a$. Slab 1 has a dielectric constant of 2, and slab 2 has a dielectric constant of 1.5. The free charge density on the top plate is σ and on the bottom plate $-\sigma$. Draw a figure showing the mentioned situation and find the following.
 - (a) Find the electric displacement D in each slab.
 - (b) Find the electric field E in each slab.
 - (c) Find the polarization P in each slab.
 - (d) Find the potential difference between the plates.
 - (e) Find the location and amount of all bound charge.
 - (f) Now that you know all the charge (free and bound), recalculate the field in each slab, and confirm your answer to (b).

4. A cube has 5 sides grounded. The sixth side insulated from the others is held at a potential V_0 . What is the potential at the center of the cube? Why?
5. The inside of a grounded spherical metal shell (inner radius R_1 and outer radius R_2) is filled with space charge of uniform charge density ρ . Find the electrostatic energy of the system? Find the potential at the center?
6. A sphere of radius R_1 has charge density ρ uniform within its volume, except for a small spherical hollow region of radius R_2 located at distance a from the center.
 - (a) Find the field $\vec{E}$ at the center of the hollow sphere.
 - (b) Find the potential ϕ at the same point.